Primer Punto

servidor1

actviamos el forwarding

```
vagrant@servidor1:~$ sudo vim /etc/ufw/sysctl.conf
vagrant@servidor1:~$
```

```
servidor1                    servidor2

#
# Configuration file for setting network var
gs
# override /etc/sysctl.conf and /etc/sysctl.
# /etc/sysctl.conf, please adjust IPT_SYSCTL
# Documentation/networking/ip-sysctl.txt in
# information.
#

# Uncomment this to allow this host to route
net/ipv4/ip_forward=1
#net/ipv6/conf/default/forwarding=1
#net/ipv6/conf/all/forwarding=1
```

permitimos reenvio en ufw

```
vagrant@servidor1:~$ sudo vim /etc/default/ufw
```

```
servidor1                                   servidor2
# /etc/default/ufw
#

# Set to yes to apply rules to support IPv6 (no means onl
# accepted). You will need to 'disable' and then 'enable'
# the changes to take affect.
IPV6=yes

# Set the default input policy to ACCEPT, DROP, or REJECT
if
# you change this you will most likely want to adjust you
DEFAULT_INPUT_POLICY="DROP"

# Set the default output policy to ACCEPT, DROP, or REJEC
 if
# you change this you will most likely want to adjust you
DEFAULT_OUTPUT_POLICY="ACCEPT"

# Set the default forward policy to ACCEPT, DROP or REJEC
t
# if you change this you will most likely want to adjust
DEFAULT_FORWARD_POLICY="ACCEPT"
```

agregamos las reglas NAT/DNAT

```
vagrant@servidor1:~$ sudo vim /etc/ufw/before.rules
```

```
servidor1                                   servidor2
#
# rules.before
#
# Rules that should be run before the ufw command line added rules. Custom
# rules should be added to one of these chains:
#   ufw-before-input
#   ufw-before-output
#   ufw-before-forward
#
*nat
:PREROUTING ACCEPT [0:0]
:POSTROUTING ACCEPT [0:0]

-A PREROUTING -p tcp -d 192.168.50.3 --dport 21 -j DNAT --to-destination 192
.168.50.2:21
-A PREROUTING -p tcp -d 192.168.50.3 --dport 50000:50010 -j DNAT --to-destin
ation 192.168.50.2:50000-50010

-A POSTROUTING -p tcp -d 192.168.50.2 --dport 21 -j MASQUERADE
-A POSTROUTING -p tcp -d 192.168.50.2 --dport 50000:50010 -j MASQUERADE

COMMIT
# Don't delete these required lines, otherwise there will be errors
*filter
:ufw-before-input - [0:0]
```

comenzamos modificando el servidor1 con ufw para este punto

```
servidor1                                   servidor2
vagrant@servidor1:~$ sudo apt update
Hit:1 http://us.archive.ubuntu.com/ubuntu jammy
Get:2 http://security.ubuntu.com/ubuntu jammy-s
Get:3 http://us.archive.ubuntu.com/ubuntu jammy
Get:4 http://us.archive.ubuntu.com/ubuntu jammy
Get:5 http://security.ubuntu.com/ubuntu jammy-s
3,067 kB]
Get:6 http://us.archive.ubuntu.com/ubuntu jammy
```

instalamos el ufw

```
vagrant@servidor1:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 111 not upgraded.
vagrant@servidor1:~$
```

- **deny incoming**: cualquier conexión que quiera entrar será bloqueada, salvo que tú la permitas explícitamente.
- **allow outgoing**: la máquina sí podrá iniciar conexiones hacia afuera.

```
vagrant@servidor1:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
vagrant@servidor1:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
vagrant@servidor1:~$
```

abrimos el puerto de ssh

```
vagrant@servidor1:~$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
vagrant@servidor1:~$
```

tambien el rango

```
vagrant@servidor1:~$ sudo ufw allow 50000:50010/tcp
Rules updated
Rules updated (v6)
vagrant@servidor1:~$
```

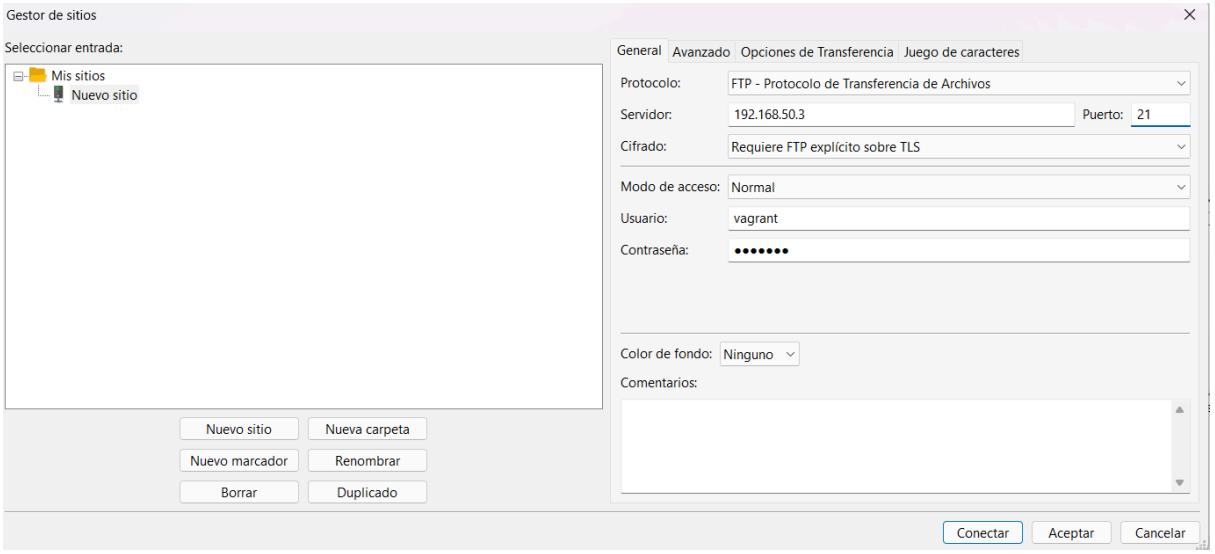
activamos el ufw y vemos las reglas

```
vagrant@servidor1:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)?
y
Firewall is active and enabled on system startup
vagrant@servidor1:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
50000:50010/tcp            ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
50000:50010/tcp (v6)       ALLOW IN    Anywhere (v6)

vagrant@servidor1:~$
```

probamos conectarnos desde el filezila



y no, nos deja para este punto de la prueba esta bien

```
Estado:    Conectando a 192.168.50.3:21...
Estado:    Falló intento de conexión con "EACCES - Permiso denegado".
Error:     No se pudo conectar al servidor
Estado:    Esperando para reintentar...
Estado:    Conectando a 192.168.50.3:21...
Estado:    Falló intento de conexión con "EACCES - Permiso denegado".
Error:     No se pudo conectar al servidor
```

Ahora si abrimos el puerto 21 y verificamos

```
vagrant@servidor1:~$ sudo ufw allow 21/tcp
Rule added
Rule added (v6)
vagrant@servidor1:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
50000:50010/tcp            ALLOW IN    Anywhere
21/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
50000:50010/tcp (v6)       ALLOW IN    Anywhere (v6)
21/tcp (v6)                ALLOW IN    Anywhere (v6)

vagrant@servidor1:~$
```

Ahora vamos a modificar el servidor2

```
servidor1                                   servidor2
vagrant@servidor2:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InR
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu jammy-updates In
Get:4 http://us.archive.ubuntu.com/ubuntu jammy-backports
Get:5 http://security.ubuntu.com/ubuntu jammy-security/mai
3,067 kB]
Get:6 http://us.archive.ubuntu.com/ubuntu jammy-updates/ma
[3,328 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/mai
437 kB]
```

instalamos el vsftpd

```
servidor1                                   servidor2
vagrant@servidor2:~$ sudo apt install vsftpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ssl-cert
The following NEW packages will be installed:
  ssl-cert vsftpd
0 upgraded, 2 newly installed, 0 to remove and 111 not upgraded.
Need to get 140 kB of archives.
After this operation, 391 kB of additional disk space will be us
Get:1 http://us.archive.ubuntu.com/ubuntu jammy/main amd64 ssl-c
2 [17.4 kB]
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd
```

creamos de nuevo los certificados

```
vagrant@servidor2:~$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:204
8 \
-keyout /etc/ssl/private/vsftpd-selfsigned.key \
-out /etc/ssl/certs/vsftpd-selfsigned.crt
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........+.+..+....+...+......+...........+....+............+..+.+.....+.+...
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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:CO
State or Province Name (full name) [Some-State]:Valle del Cauca
Locality Name (eg, city) []:Cali
Organization Name (eg, company) [Internet Widgits Pty Ltd]:University
Organizational Unit Name (eg, section) []:Ciencias Computacionales
Common Name (e.g. server FQDN or YOUR name) []:192.168.50.2
Email Address []:admin@servicios.com
vagrant@servidor2:~$
```

verificamos los archivos de las contraseñas

```
vagrant@servidor2:~$ sudo ls -l /etc/ssl/private/vsftpd-selfsigned.key
-rw------- 1 root root 1704 Mar 31 17:16 /etc/ssl/private/vsftpd-selfsigned.
key
vagrant@servidor2:~$ ls -l /etc/ssl/certs/vsftpd-selfsigned.crt
-rw-r--r-- 1 root root 1521 Mar 31 17:18 /etc/ssl/certs/vsftpd-selfsigned.cr
t
vagrant@servidor2:~$
```

configuramos el vsftpd para usar el certificado

```
vagrant@servidor2:~$ sudo vim /etc/vsftpd.conf
vagrant@servidor2:~$
```

```
listen=YES
listen_ipv6=NO

anonymous_enable=NO
local_enable=YES
write_enable=YES
local_umask=022

chroot_local_user=YES
allow_writeable_chroot=YES

ssl_enable=YES
allow_anon_ssl=NO
force_local_data_ssl=YES
force_local_logins_ssl=YES
ssl_tlsv1=YES
ssl_sslv2=NO
ssl_sslv3=NO
require_ssl_reuse=NO

rsa_cert_file=/etc/ssl/certs/vsftpd-selfsigned.crt
rsa_private_key_file=/etc/ssl/private/vsftpd-selfsigned.key

pasv_enable=YES
pasv_min_port=50000
pasv_max_port=50010
pasv_address=192.168.50.3
~
```

Qué hace esa parte

- activa FTPS con TLS
- obliga a que login y datos vayan cifrados
- usa el certificado autofirmado recién creado
- define el rango pasivo `50000-50010`

Reiniciamos y verificamos

```
vagrant@servidor2:~$ sudo systemctl restart vsftpd
vagrant@servidor2:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/lib/systemd/system/vsftpd.service; enabled; vendor pr>
     Active: active (running) since Tue 2026-03-31 17:34:25 UTC; 10s ago
    Process: 2234 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=ex>
   Main PID: 2236 (vsftpd)
      Tasks: 1 (limit: 2219)
     Memory: 1.6M
        CPU: 39ms
     CGroup: /system.slice/vsftpd.service
             └─2236 /usr/sbin/vsftpd /etc/vsftpd.conf

Mar 31 17:34:25 servidor2 systemd[1]: Starting vsftpd FTP server...
Mar 31 17:34:25 servidor2 systemd[1]: Started vsftpd FTP server.
lines 1-13/13 (END)
vagrant@servidor2:~$ sudo ss -tulnp | grep :21
tcp   LISTEN 0      32           0.0.0.0:21        0.0.0.0:*    users:(("vs
ftpd",pid=2236,fd=3))
vagrant@servidor2:~$
```

creamos un nuevo usuario para las pruebas

```
vagrant@servidor2:~$ sudo adduser juanpalacios
Adding user `juanpalacios' ...
Adding new group `juanpalacios' (1001) ...
Adding new user `juanpalacios' (1001) with group `juanpalacios' ...
Creating home directory `/home/juanpalacios' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for juanpalacios
Enter the new value, or press ENTER for the default
        Full Name []: juan diego
        Room Number []: 310658
        Work Phone []: 310658
        Home Phone []: 310658
        Other []:
Is the information correct? [Y/n] y
```

creamos un archivos de prueba

```
vagrant@servidor2:~$ sudo -u juanpalacios touch /home/juanpalacios/archivo_p
rueba.txt
vagrant@servidor2:~$ ls -l /home/juanpalacios/
ls: cannot open directory '/home/juanpalacios/': Permission denied
vagrant@servidor2:~$ sudo ls -l /home/juanpalacios/
total 0
-rw-rw-r-- 1 juanpalacios juanpalacios 0 Mar 31 17:58 archivo_prueba.txt
vagrant@servidor2:~$
```

probamos desde filezilla

Certificado desconocido

El certificado del servidor es desconocido. Por favor, examine cuidadosamente el certificado para asegurarse de que se puede confiar en el servidor.

Compare la huella digital que se muestra con la huella digital del certificado que tiene recibido de su administrador de servidor o proveedor de alojamiento de servidor.

Certificado

Vista previa

Huella digital (SHA-256):	1c:3f:40:9c:fd:d8:a0:58:a8:fc:a0:e7:dd:6f:b3:27:e4:32:8c:7e:9a:b0:fd:c8:cd:da:c0:a4:3e:d3:37:15
Huella digital (SHA-1):	5a:7b:73:e9:e9:2e:22:a9:7d:8f:d7:12:85:81:d8:bd:28:bf:84:fa
Período de validez:	De 31 mar. 2026 12:18:42 a 31 mar. 2027 12:18:42

Asunto

Nombre común:	192.168.50.2
Organización:	University
Unidad:	Ciencias Computacionales
País:	CO
Estado o provincia:	Valle del Cauca
Localidad:	Cali
Correo electrónico:	admin@servicios.com

Editor

Igual que el asunto, el certificado está autofirmado

Detalles

De serie:	33:77:75:1a:a5:94:c2:60:45:d0:02:96:c8:a0:93:b8:e8:32:a4:38
Algoritmo de clave pública:	RSA con 2048 bits
Algoritmo de firma:	RSA-SHA256

Detalles de la sesión

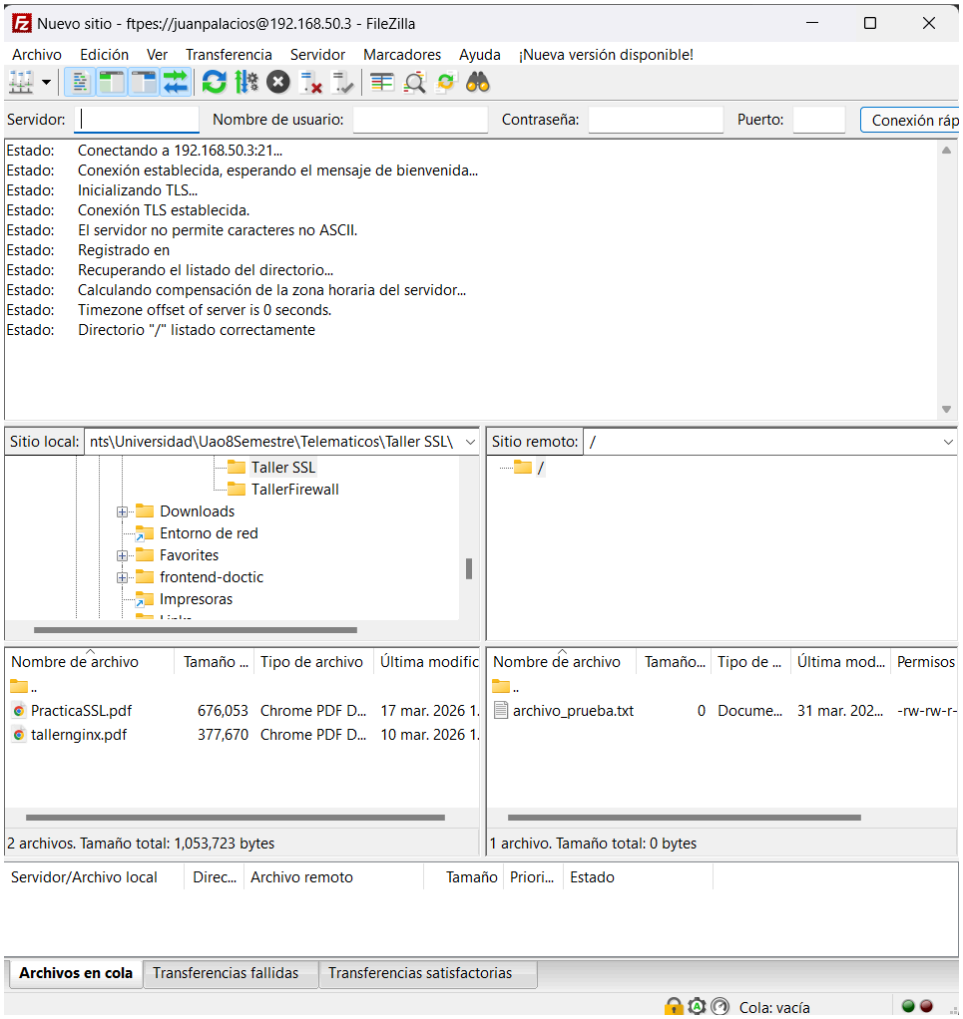
Sitio:	192.168.50.3:21		
Protocolo:	TLS1.3	Cifrado:	AES-256-GCM
Intercambio de clave:	ECDHE-SECP256R1-RSA-PSS-RSAE-SHA384	Mac:	AEAD

¿Confiar en el certificado del servidor y continuar con la conexión?

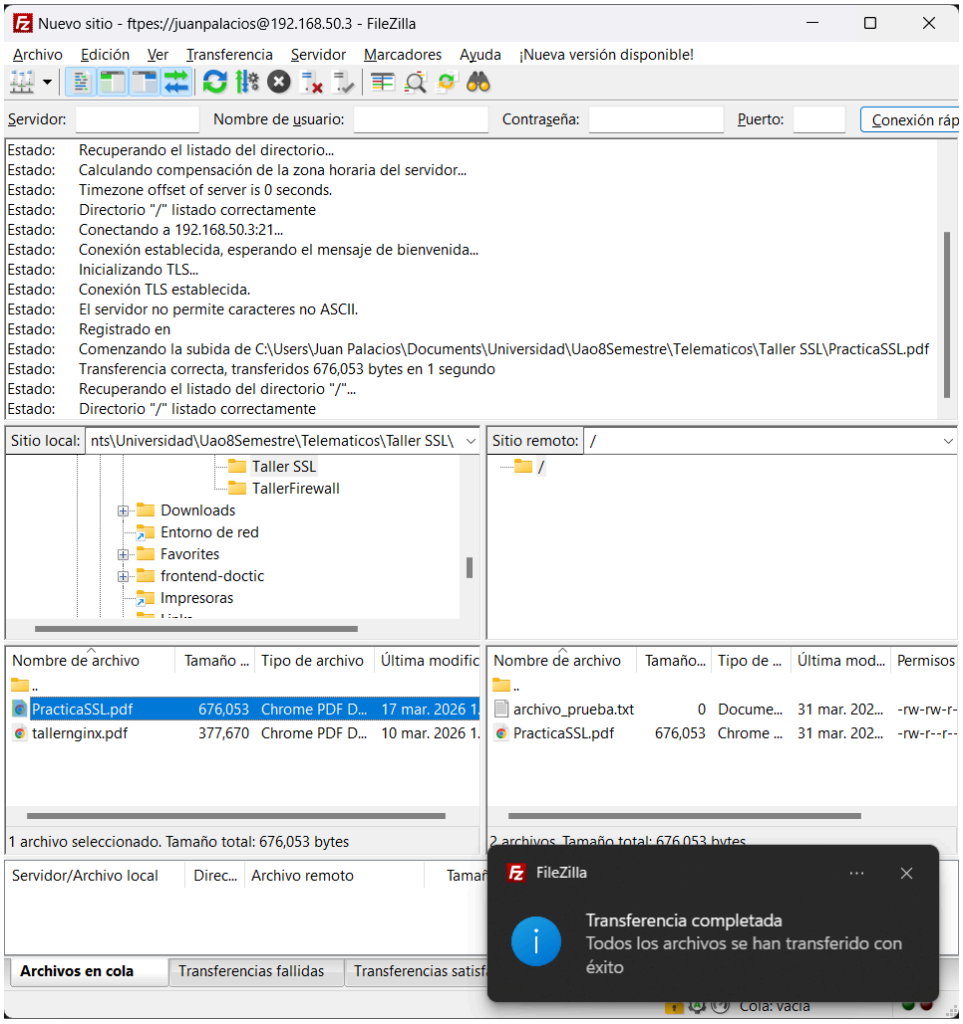
☐ Confiar siempre en este certificado en futuras sesiones.

☐ Confiar en este certificado sobre los nombres de servidor alternativos de la lista.

Aceptar Cancelar



subimos un archivo



eliminamos para hacer la prueba del punto 1.3

```
vagrant@servidor1:~$ sudo ufw delete 3
Deleting:
 allow 21/tcp
Proceed with operation (y|n)? y
Rule deleted
vagrant@servidor1:~$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 50000:50010/tcp            ALLOW IN    Anywhere
[ 3] 22/tcp (v6)                ALLOW IN    Anywhere (v6)
[ 4] 50000:50010/tcp (v6)       ALLOW IN    Anywhere (v6)
[ 5] 21/tcp (v6)                ALLOW IN    Anywhere (v6)

vagrant@servidor1:~$ sudo ufw delete 5
Deleting:
 allow 21/tcp
Proceed with operation (y|n)? y
Rule deleted (v6)
vagrant@servidor1:~$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 50000:50010/tcp            ALLOW IN    Anywhere
[ 3] 22/tcp (v6)                ALLOW IN    Anywhere (v6)
[ 4] 50000:50010/tcp (v6)       ALLOW IN    Anywhere (v6)

vagrant@servidor1:~$ |
```